

Anomaly Rescue Mission Classroom Run Kit

TOK Cognitive Science Activity Bank • Natural Sciences • 30-40 min

Category	Natural Sciences	Time	30-40 min
Difficulty	Medium	ID	anomaly-rescue-mission

Big idea: Anomalies are not automatically errors; they are tests of method, theory, and judgement.

One-Minute Teacher Brief

Students receive a dataset with one anomalous result and decide whether to reject, explain, repeat, or revise the model.

Student Prompt

Inspect Anomaly Rescue Mission Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Anomaly Rescue Mission Stimulus A	Student-facing stimulus 1: Plant growth mostly increases with light exposure, except one low-light plant grows unusually tall. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Anomaly Rescue Mission Stimulus B	Student-facing stimulus 2: Reaction time improves with practice, except one trial suddenly becomes much slower. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Use this for comparison, a second round, or a partner challenge.
Anomaly Rescue Mission Reveal / Twist	Student-facing stimulus 3: Reaction time improves with practice, except one trial suddenly becomes much slower. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Teacher reveal: connect the changed response to the big idea — Anomalies are not automatically errors; they are tests of method, theory, and judgement.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Anomaly Rescue Mission Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Anomaly Rescue Mission Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.

#	Question for students	Teacher key / cue
6	When should anomalous evidence change a theory?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Natural Sciences.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how anomalies are not automatically errors; they are tests of method, theory, and judgement.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: When should anomalous evidence change a theory?

Setup Checklist

- Use a dataset simple enough for students to understand quickly but ambiguous enough to support different responses.
- Make the anomaly plausible as either error or discovery.

Example Stimuli or Materials

- Plant growth mostly increases with light exposure, except one low-light plant grows unusually tall.
- Reaction time improves with practice, except one trial suddenly becomes much slower.

Resources To Use

- Resource pack: <resources/anomaly-rescue-mission/index.html>
- Card deck CSV: <resources/anomaly-rescue-mission/cards.csv>
- Visual prompt: <resources/anomaly-rescue-mission/visual-prompt.svg>
- Anomaly decision board: discard, repeat, investigate, revise model.
- Evidence-needed cards: check instrument, repeat measurement, inspect conditions, compare theory.

Run Sequence

1. Give groups a simple model and a dataset that mostly fits it.
2. Students identify the anomalous result and list possible explanations.
3. Groups choose a response: discard, repeat, investigate, or revise the model.
4. Groups defend what evidence would justify their response.
5. Debrief how scientists should treat results that do not fit expectation.

Debrief Moves

- Knowledge question: When should anomalous evidence change a theory?
- How can expectations help and hinder scientific judgement?
- What is the difference between ignoring an anomaly and responsibly investigating it?

Teacher Quick Script

- An anomaly is a question mark, not automatically a mistake or a breakthrough.
- What evidence would justify your next move?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...